
NUMERICAL QUESTIONS ON STRESS, STRAIN & PIEZORESISTANCE

Q1. A mild steel wire of length 2 m and diameter 1 mm is subjected to a tensile force of 50 N.

If Young's modulus for steel is $E = 2\times10^{11}$ Pa, determine:

1. The stress in the wire.
2. The strain produced.
3. The extension in the wire.

Q2. A brass rod of cross-sectional area 10 mm^2 and length 0.5 m is stretched by 0.25 mm under a load. Find the applied stress and check whether it exceeds the elastic limit if the yield strength of brass is 2.5×10^{8} Pa. Use $E = 1\times10^{11}$ Pa.

Q3. A steel cube of side 10 cm is subjected to a uniform pressure of 50 MPa. Given bulk modulus $K = 1.6\times10^{11}$ Pa, calculate the change in volume of the cube.

Q4. A steel rod (length = 0.5 m, diameter = 15 mm) is under a tensile stress of 120 MPa. Given Poisson's ratio $\nu = 0.3$ and $E = 2\times10^{11}$ Pa:
Find axial strain, lateral strain, change in diameter.

Q5. At a point inside a component, stress components are: $\sigma_x = 100$ MPa, $\sigma_y = 40$ MPa, $\sigma_z = 20$ MPa. Given E = 200 GPa and $\nu = 0.25$, calculate:

1. Strain in the x-direction
2. Volumetric strain.

Q6. A Constantan (Cu-Ni alloy) strain gauge has a resistance of 120 Ω and a gauge factor GF = 2.0. If it experiences a strain of 500 με, calculate change in resistance ΔR.

Q7. A lightly doped n-type silicon resistor has $\rho = 10\ \Omega\cdot$cm. When subjected to a stress of 100 MPa, the piezoresistive coefficient is $\pi_L = -102\times10^{-11}\ Pa^{-1}$. Compute $\Delta\rho/\rho$ and the new ρ. [Given $\Delta\rho/\rho = \pi_L\sigma$]

Q8. An NMOS transistor channel (length = 1 μm, width = 10 μm) experiences a longitudinal stress of **200 MPa** due to packaging. If the **electron mobility** changes according to $\Delta\mu/\mu = \pi_L\sigma$ where $\pi_L = -30\times10^{-11}$ Pa^{-1}, compute fractional mobility change and comment whether the current increases or decreases.

$\Delta\mu/\mu = -30\times10^{-11}\times2\times10^{8} = -6\times10^{-2}$ ⇒ current decreases

Q9. A silicon piezoresistor of length 1 mm, area 50 μm × 10 μm and resistivity 5×10^{-3} Ω·m is subjected to a tensile strain of 1×10^{-4}.

$R = \frac{5\times10^{-3}\times10^{-3}}{5\times10^{-10}}$

$100 = \frac{\Delta R/R}{10^{-4}} \Rightarrow \frac{\Delta R}{R} = 10^{-2} \rightarrow \Delta R = 100\Omega$

$R_{new} = 10^{4}(1.01)$

The gauge factor GF = 100. [GF = $(\Delta R/R)/\epsilon$]

Compute the initial resistance R_0, the change in resistance ΔR, and new resistance R.

Q10. A p-type silicon resistor has a temperature coefficient of resistivity α_T = 7×10^{-4}/°C and a piezoresistive coefficient $\pi_L = 70\times10^{-11}$ Pa^{-1}, For a stress of 50 MPa and a temperature rise of 10°C, compute the total fractional change in resistivity:

$\Delta\rho/\rho = \alpha_T\,\Delta T + \pi_L\sigma$ $\quad 7\times10^{-4}(10) + 70\times10^{-11}\times5\times10^{7} = 7\times10^{-3} + 35\times10^{-3} = 42\times10^{-3}$